POWER BI MODULE END PROJECT REPORT

AI-POWERED FLEET PERFORMANCE & DELIVERY EFFICIENCY DASHBOARD

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Power Bi File : Gayathri Power Bi Module End Project

1. Project Purpose

The purpose of this project is to analyse delivery operations using Power BI to understand fuel usage, delivery efficiency, cost per km, and vehicle performance. The insights help management improve fleet utilization, reduce maintenance cost, and increase on-time delivery rate.

2. Project Objective

- Clean and combine trip-level data and vehicle details
- Fix missing values
- Build relationships using Vehicle_ID
- Create DAX measures for performance metrics
- Build dashboards for delivery performance
- Use AI visuals to explore influencers and patterns
- Generate insights for fleet optimization

3. Data Cleaning Steps

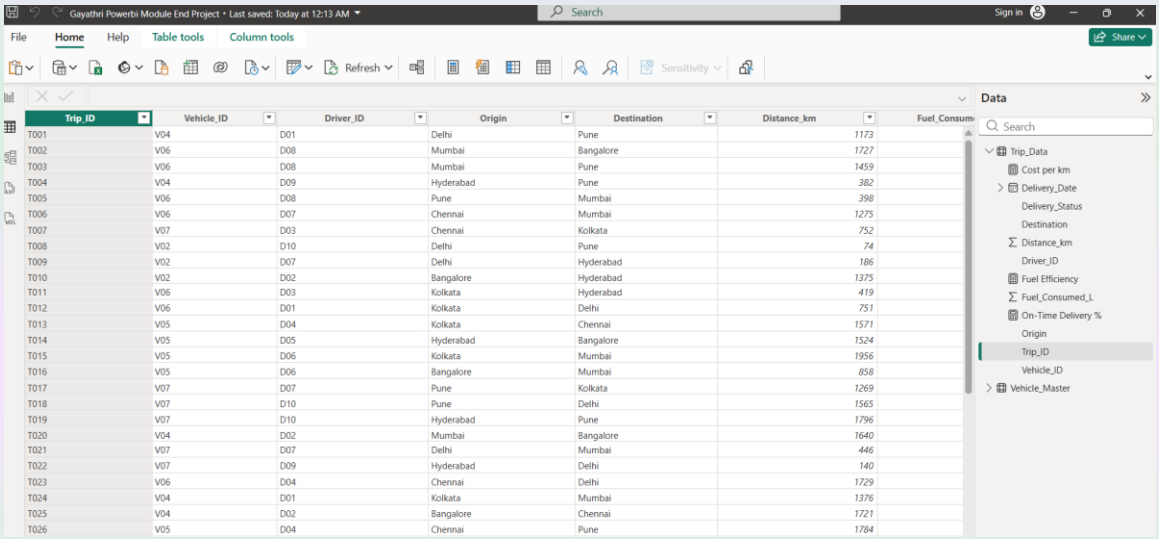
Trip_Data:

- Distance: Checked numeric type, removed nulls
- Fuel_Consumed: Missing values replaced using mean imputation
- Maintenance_Cost: Verified numeric
- Status: Standardized (“On-Time”, “Late”)
- Delivery_Time: Outliers checked
- Delivery_Date: Converted to date format
- Vehicle_ID: Used for relationship creation

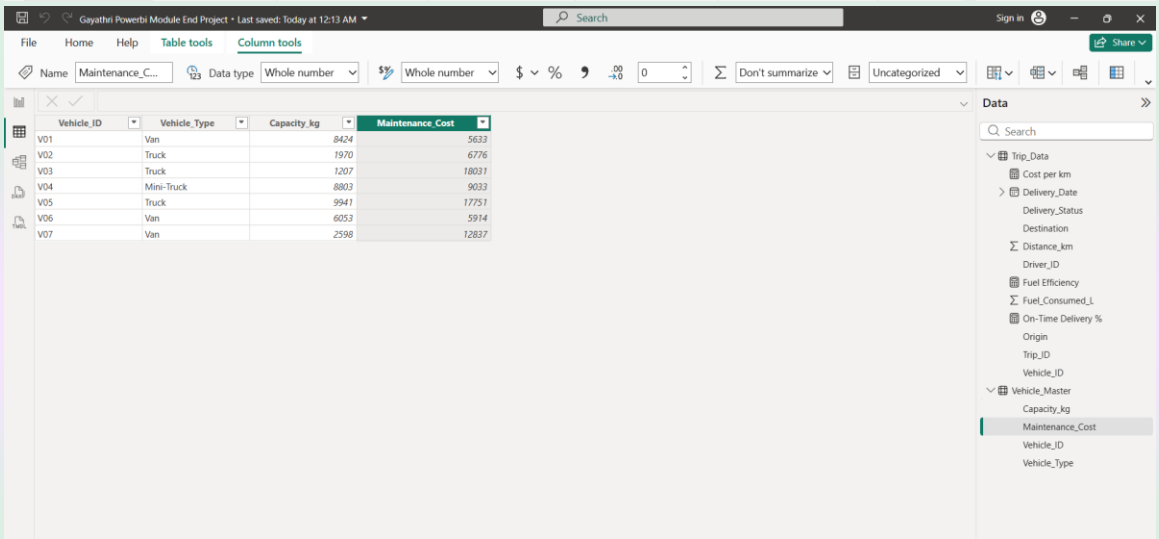
Vehicle_Master:

- Vehicle_ID: Duplicates removed
- Vehicle_Type: Normalized text
- Capacity: Verified numeric
- Registration No: Checked for missing

Here the Screenshot for after data cleaning below:



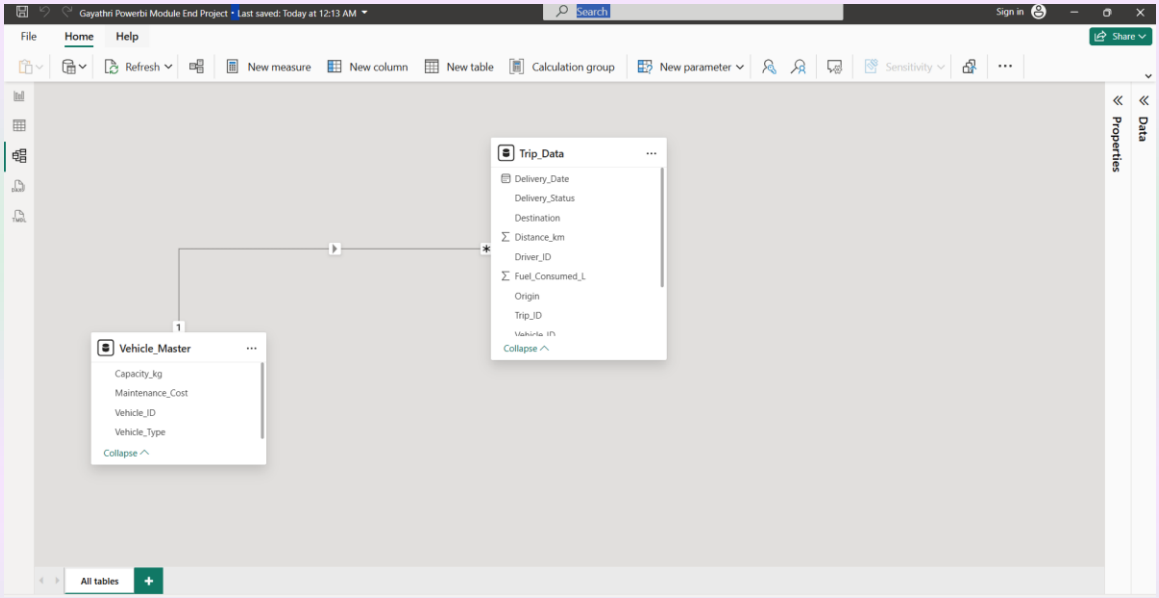
Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consum
T001	V04	D01	Delhi	Pune	1173	
T002	V06	D08	Mumbai	Bangalore	1727	
T003	V06	D08	Mumbai	Pune	1459	
T004	V04	D09	Hyderabad	Pune	382	
T005	V06	D08	Pune	Mumbai	398	
T006	V06	D07	Chennai	Mumbai	1275	
T007	V07	D03	Chennai	Kolkata	752	
T008	V02	D10	Delhi	Pune	74	
T009	V02	D07	Delhi	Hyderabad	186	
T010	V02	D02	Bangalore	Hyderabad	1375	
T011	V06	D03	Kolkata	Hyderabad	419	
T012	V06	D01	Kolkata	Delhi	751	
T013	V05	D04	Kolkata	Chennai	1571	
T014	V05	D05	Hyderabad	Bangalore	1524	
T015	V05	D06	Kolkata	Mumbai	1956	
T016	V05	D06	Bangalore	Mumbai	858	
T017	V07	D07	Pune	Kolkata	1269	
T018	V07	D10	Pune	Delhi	1565	
T019	V07	D10	Hyderabad	Pune	1796	
T020	V04	D02	Mumbai	Bangalore	1640	
T021	V07	D07	Delhi	Mumbai	446	
T022	V07	D09	Hyderabad	Delhi	140	
T023	V06	D04	Chennai	Delhi	1729	
T024	V04	D01	Kolkata	Mumbai	1376	
T025	V04	D02	Bangalore	Chennai	1721	
T026	V05	D04	Chennai	Pune	1784	



Vehicle_ID	Vehicle_Type	Capacity_kg	Maintenance_Cost
V01	Van	8424	5633
V02	Truck	1970	6776
V03	Truck	1207	18031
V04	Mini-Truck	8803	9033
V05	Truck	9941	17751
V06	Van	6053	5914
V07	Van	2598	12837

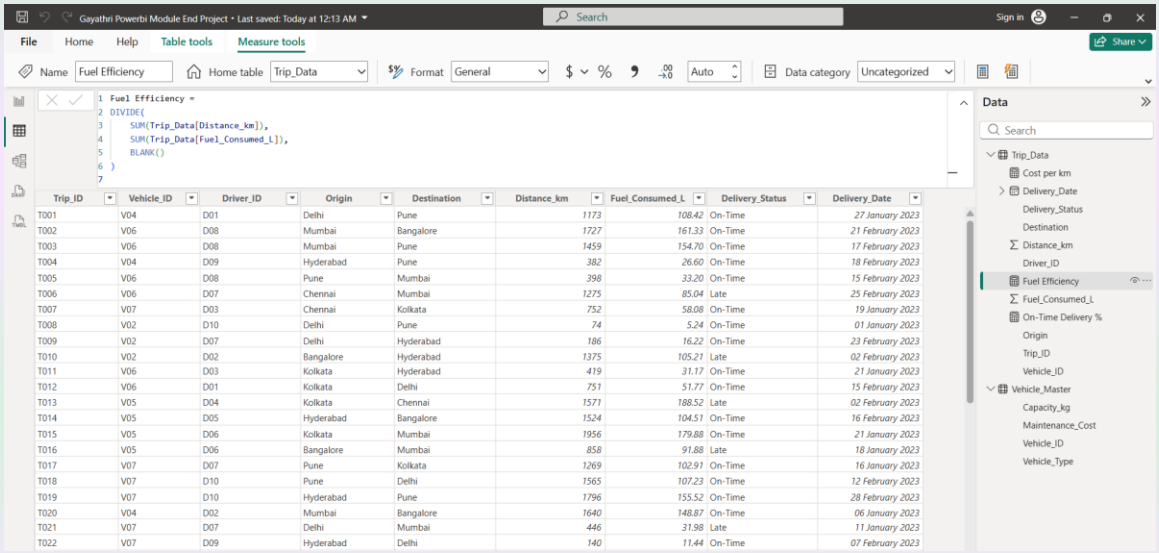
4. Data Modeling

Relationship: Trip_Data[Vehicle_ID] →Vehicle_Master[Vehicle_ID]



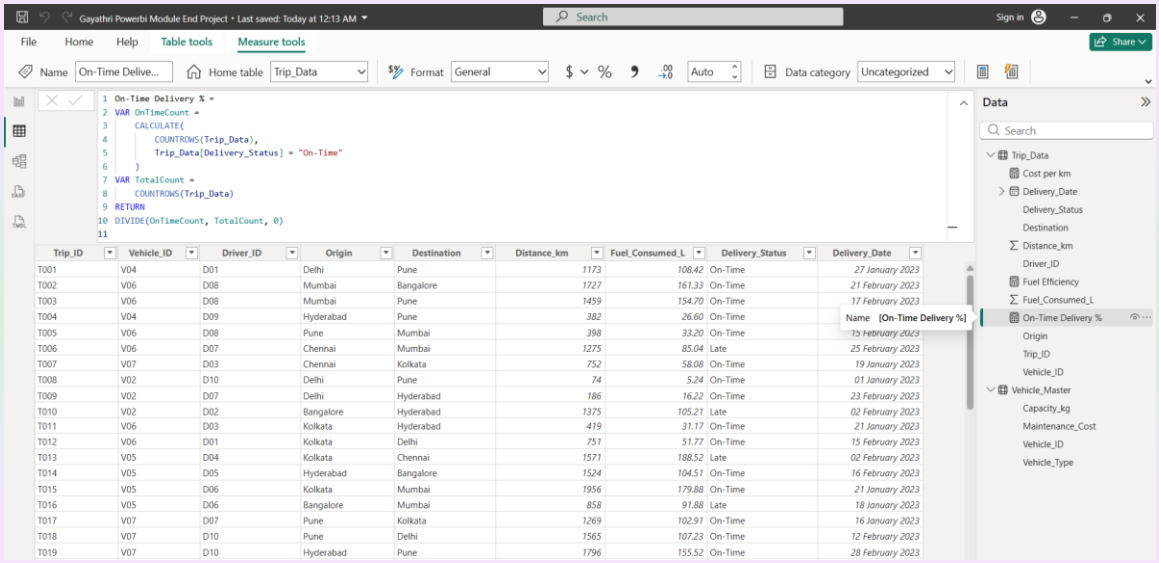
5. DAX Measures

Fuel Efficiency:

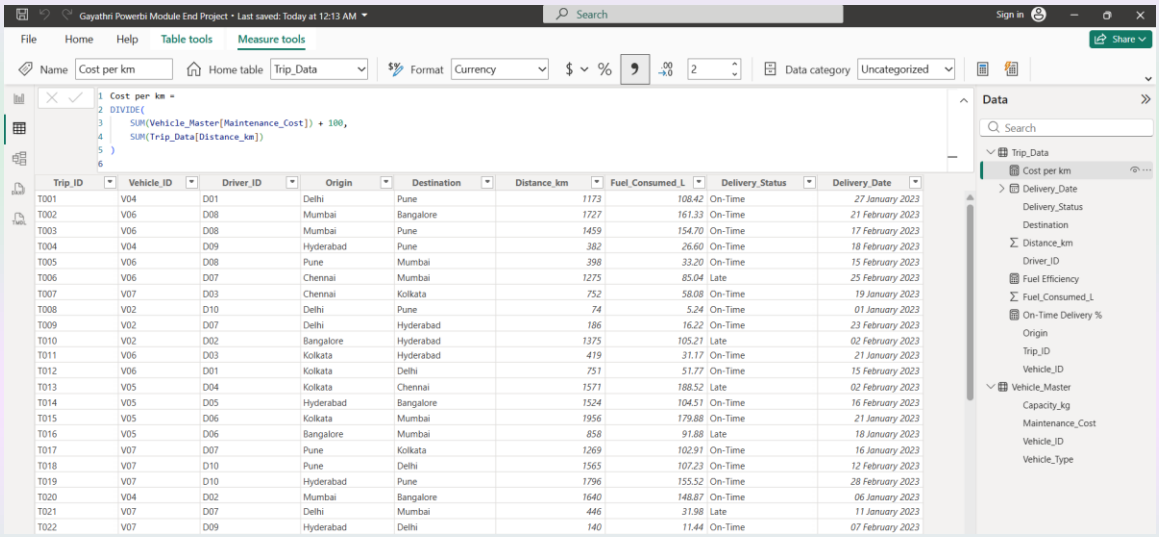


Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date
T001	V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023
T002	V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023
T003	V06	D08	Mumbai	Pune	1459	154.70	On-Time	17 February 2023
T004	V04	D09	Hyderabad	Pune	382	26.60	On-Time	18 February 2023
T005	V06	D08	Pune	Mumbai	398	33.20	On-Time	15 February 2023
T006	V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023
T007	V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023
T008	V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023
T009	V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023
T010	V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023
T011	V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023
T012	V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023
T013	V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023
T014	V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023
T015	V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023
T016	V05	D06	Bangalore	Mumbai	858	91.88	Late	18 January 2023
T017	V07	D07	Pune	Kolkata	1269	102.91	On-Time	16 January 2023
T018	V07	D10	Pune	Delhi	1565	107.23	On-Time	12 February 2023
T019	V07	D10	Hyderabad	Pune	1796	155.52	On-Time	28 February 2023
T020	V04	D02	Mumbai	Bangalore	1640	148.87	On-Time	06 January 2023
T021	V07	D07	Delhi	Mumbai	446	31.98	Late	11 January 2023
T022	V07	D09	Hyderabad	Delhi	140	11.44	On-Time	07 February 2023

On-Time Delivery %:

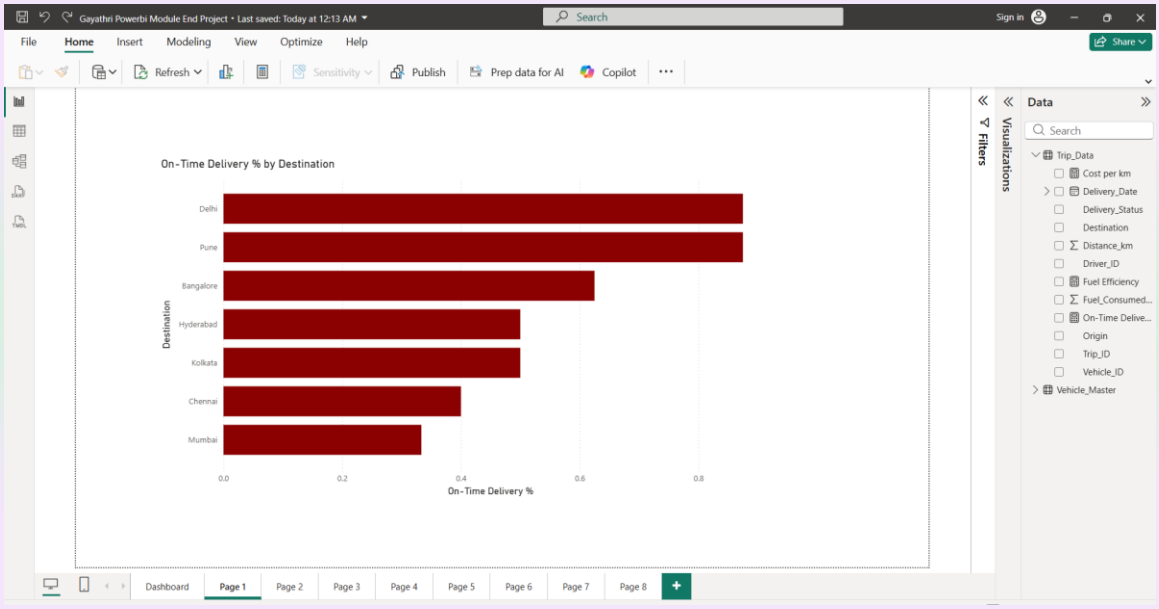


Cost per km:

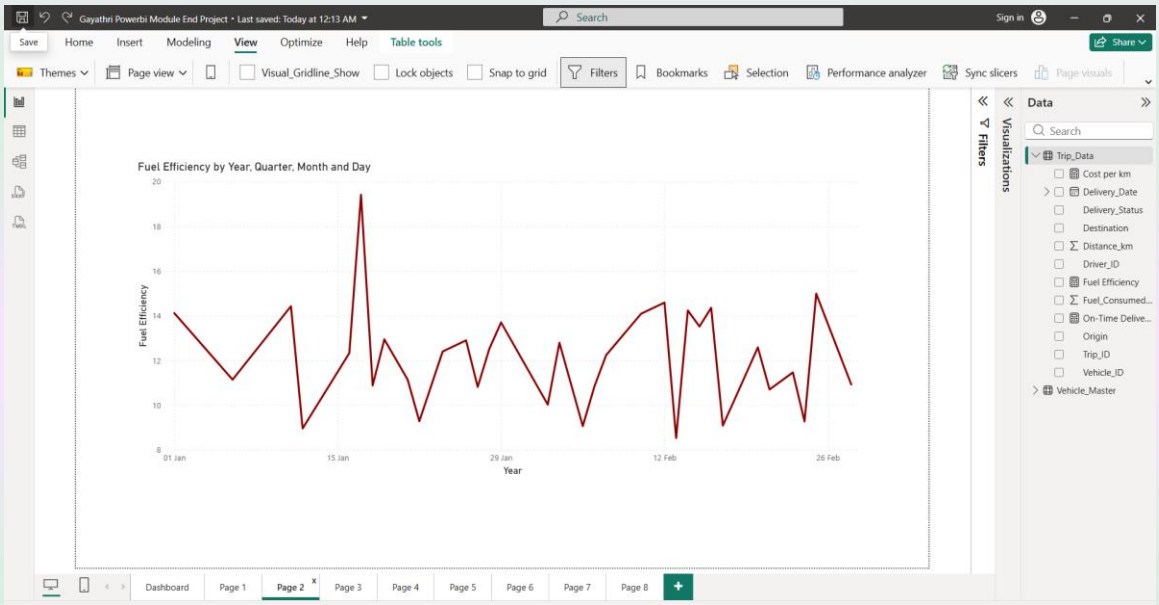


6. Visualizations

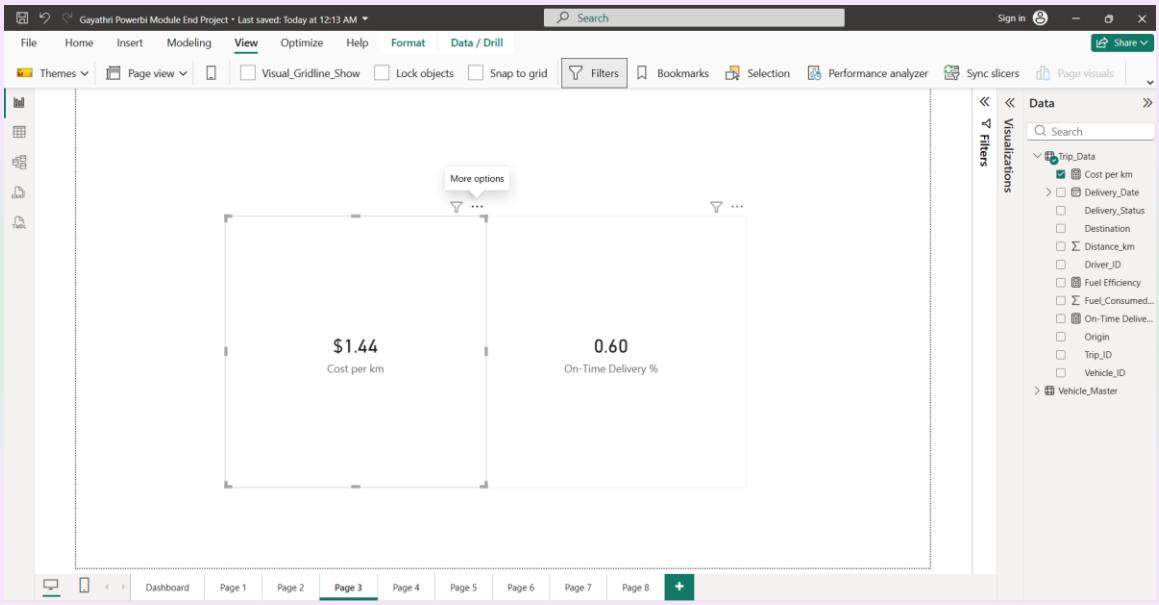
- Bar Chart: On-Time Delivery % by Destination



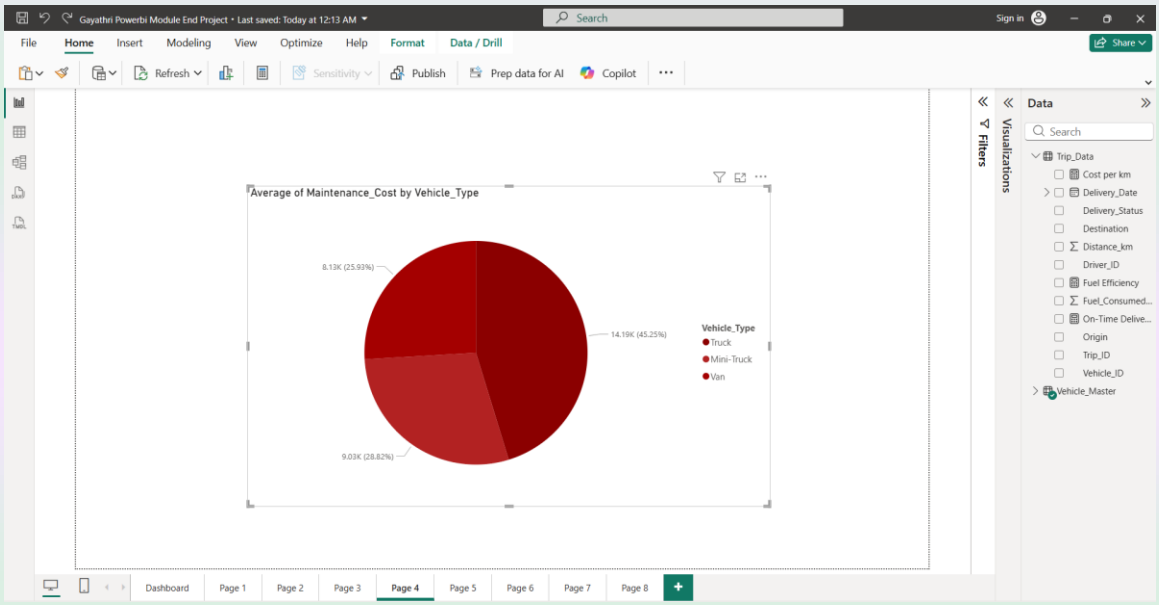
- Line Chart: Fuel Efficiency Trend by Delivery Date



- Cards: Avg Delivery Time, Avg Cost per km

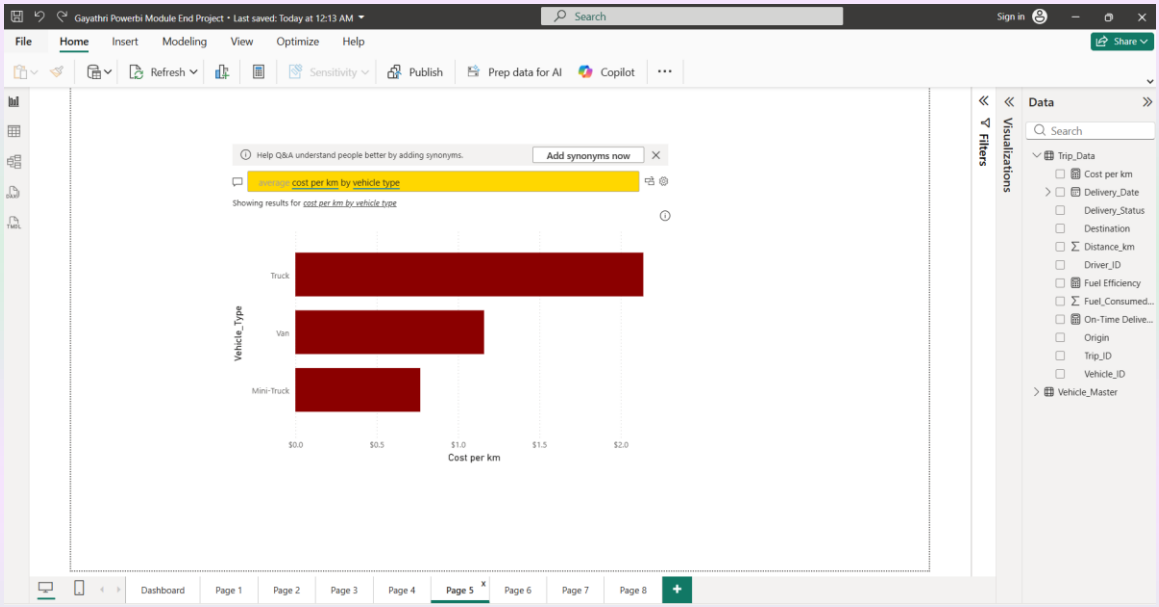


- Pie Chart: Vehicle Type vs Avg Maintenance Cost

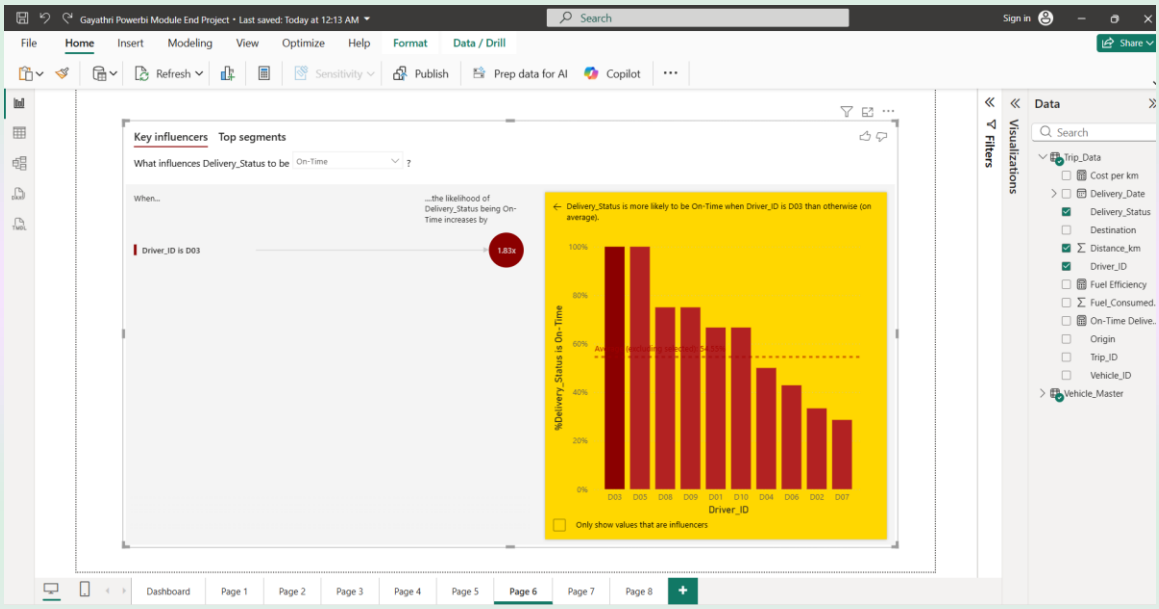


7. AI-Powered Visuals

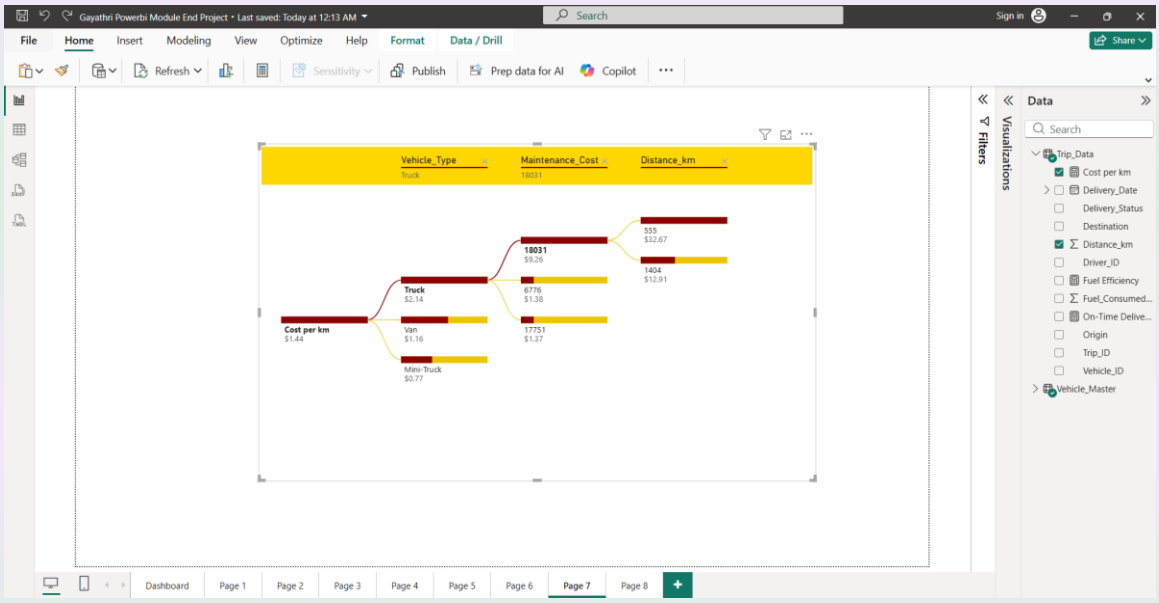
Q&A: “Average Cost per km by vehicle type?”



Key Influencers: Analyze Delivery Status by Distance_km, Vehicle_Type, Driver_ID



Decomposition Tree: Analyze Cost per km explained by Vehicle Type, Maintenance_Cost, Distance_km



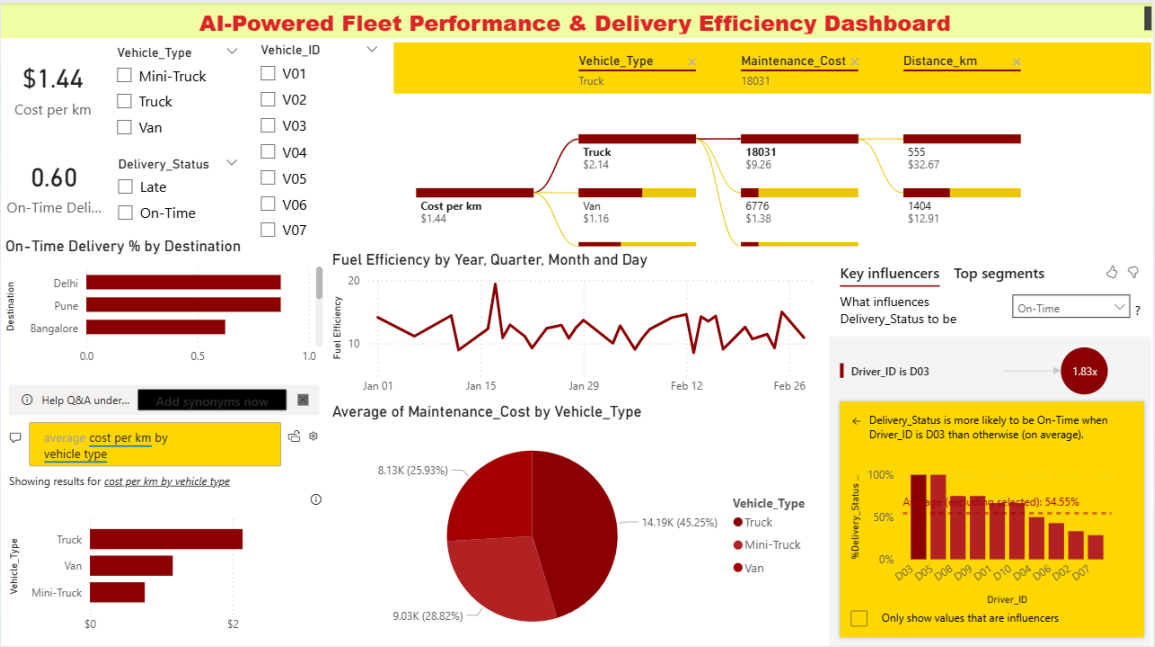
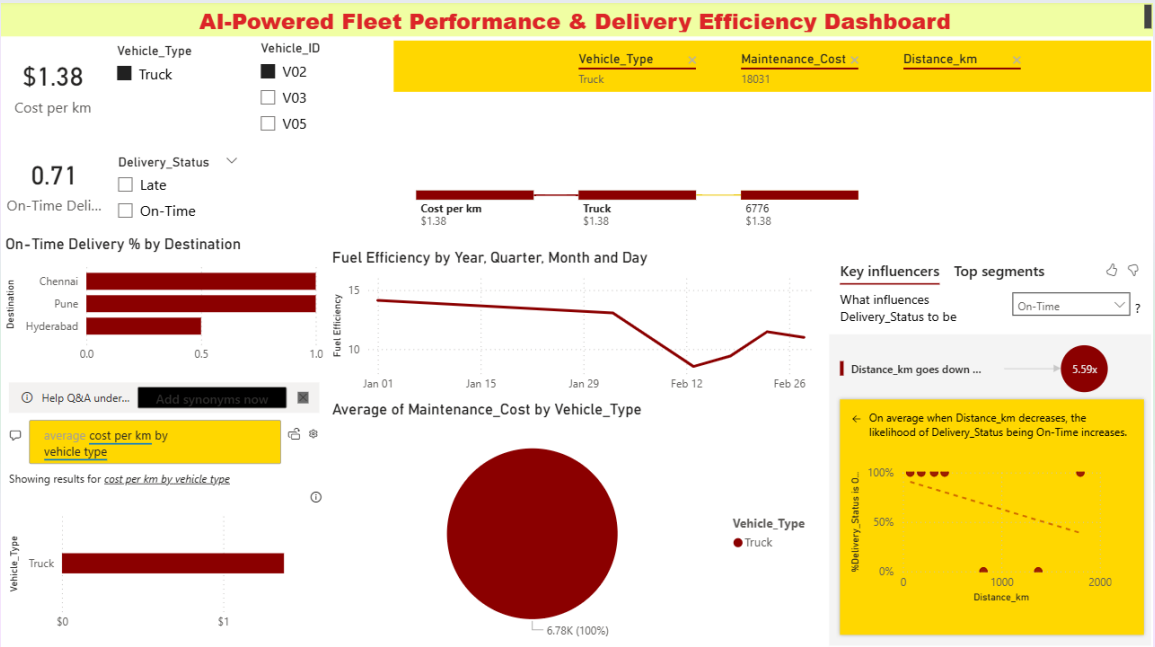
8. Final Insights & Recommendations

- Improve fuel efficiency via maintenance
- Reduce high maintenance fleet
- Optimize routes for better delivery time
- Reduce cost per km using preventive maintenance

9. Conclusion

The Power BI dashboard provides complete fleet and delivery analytics with actionable insights.

FINAL DASHBOARD SCREENSHOT(WITHOUT APPLYING SLICERS)



FINAL DASHBOARD SCREENSHOT(APPLIED SLICERS)